

6406535902835. ✓ $\langle (\frac{a_0+b_0}{2}) * (\frac{a_0+b_0}{2}) - a_0 * b_0 \leq 0 \rangle$

6406535902836. ✗ $\langle (\frac{a_0+b_0}{2}) - a_0 * b_0 > 0 \rangle$

Question Number : 47 Question Id : 6406531815631 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following sets of concrete test inputs execute all instances of symbolic execution for the method process?

Options :

6406535902837. ✗ {a = 4, b = 8}, {a = 4, b = 6}

6406535902838. ✓ {a = 6, b = 4}, {a = 0, b = 0}

6406535902839. ✗ {a = 0, b = 8}, {a = 4, b = 0}

6406535902840. ✗ {a = -1, b = 1}, {a = 4, b = 4}

AI

Section Id :	640653121908
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	6
Number of Questions to be attempted :	6
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653318016
Question Shuffling Allowed :	No

Question Number : 48 Question Id : 6406531815632 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535902841.  YES

6406535902842.  NO

Sub-Section Number :

2

Sub-Section Id :

640653318017

Question Shuffling Allowed :

No

Question Id : 6406531815633 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

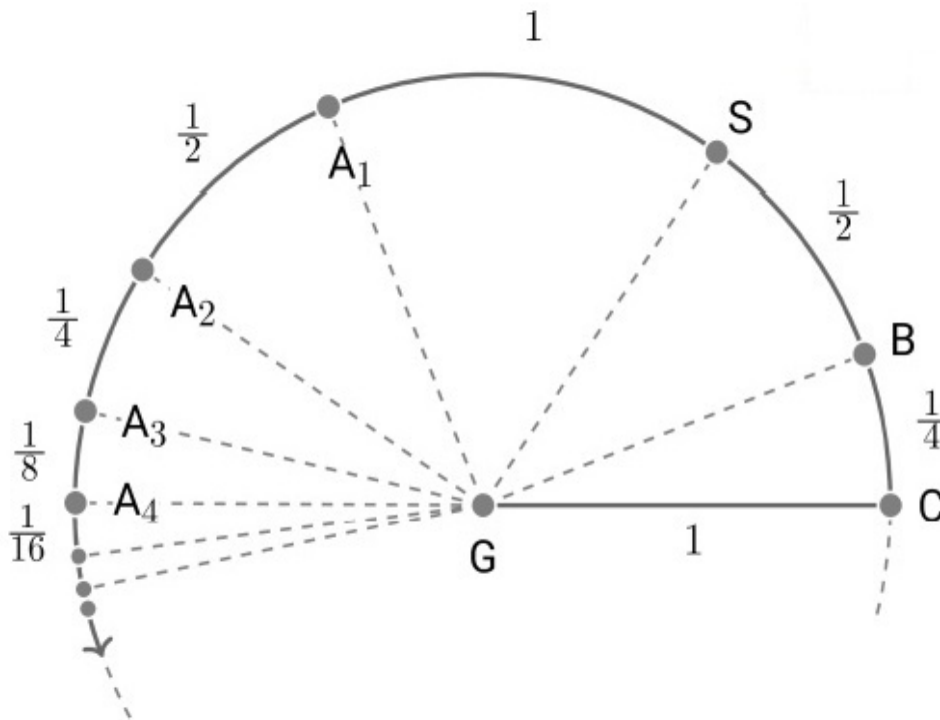
Question Numbers : (49 to 54)

Question Label : Comprehension

SEARCH

Figure shows an infinite state space with finite branching, where S is the start node and G is the goal node, and G is at the center of a unit circle, and the remaining nodes are on the unit circle.

Note: circular grid (dashed lines) is not part of the state space.



Heuristic values: $h(A_k)=1$, for k in $\{1,2,\dots\}$; $h(S)=4$; $h(B)=3$; $h(C)=2$; $h(G)=0$.

MoveGen returns neighbours in alphabetical order (A₁,A₂,...,B,C,G,S).

Tie-breaker: use alphabetical order to break ties.

Note: $(1 - a^n) / (1 - a) = 1 + a + a^2 + a^3 + \dots + a^{(n-1)}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 49 Question Id : 6406531815634 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the Depth First Search algorithm find the path to the goal?

Options :

6406535902843. ✖ Yes

6406535902844. ✔ No

6406535902845. ✖ Cannot be determined

Question Number : 50 Question Id : 6406531815635 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the Best-First Search algorithm find the path to the goal?

Options :

6406535902846. ✖ Yes

6406535902847. ✔ No

6406535902848. ✖ Cannot be determined

Question Number : 51 Question Id : 6406531815636 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the Branch and Bound algorithm find the path to the goal?

Options :

6406535902849. ✔ Yes

6406535902850. ✖ No

6406535902851. ✖ Cannot be determined

Question Number : 52 Question Id : 6406531815637 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the A* algorithm find the
path to the goal?

Options :

6406535902852. ✖ Yes

6406535902853. ✔ No

6406535902854. ✖ Cannot be determined

Question Number : 53 Question Id : 6406531815638 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the wA* (w=2) algorithm
find the path to the goal?

Options :

6406535902855. ✖ Yes

6406535902856. ✔ No

6406535902857. ✖ Cannot be determined

Question Number : 54 Question Id : 6406531815639 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the heuristic admissible in the given state space?

Options :

6406535902858. ✖ Yes

6406535902859. ✔ No

6406535902860. ✖ Cannot be determined

Sub-Section Number :

3

Sub-Section Id :

640653318018

Question Shuffling Allowed :

No

Question Id : 6406531815640 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (55 to 56)

Question Label : Comprehension

SEARCH METHODS

Based on the above topic, answer the given subquestions.

Sub questions

Question Number : 55 Question Id : 6406531815641 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following algorithms can be used to find the **most expensive path** to goal in a state space where every move is reversible and the edge costs are greater than 1?

Options :

BnB-1 (which allows cyclic expansions)
modified to refine the largest partial

6406535902861. ✖ solution.

BnB-2 (which prevents cyclic expansions)
modified to refine the largest partial
solution.

6406535902862. ✔

A* modified to select the largest f-value
node from OPEN and save pointers to
parents that lead to largest g-values.

6406535902863. ✖

6406535902864. ✖ None of these.

Question Number : 56 Question Id : 6406531815642 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

If $h(x)$ and $k(x)$ are admissible then which of the following is/are admissible?

Options :

6406535902865. ✓ $h(x)/2 + k(x)/2$

6406535902866. ✗ $h(x) + k(x)$

6406535902867. ✓ $\max(h(x), k(x))$

6406535902868. ✓ $\min(h(x), k(x))$

Sub-Section Number :

4

Sub-Section Id :

640653318019

Question Shuffling Allowed :

No

Question Id : 6406531815643 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (57 to 61)

Question Label : Comprehension

TSP

The distance matrix for 5 cities (A to E) and the sorted edge list is provided below.

For each city the distances to other cities are listed in ascending order.

For example, in row one, the distance from A to D is 50, A to E is 76 and so on.

A	D: 50	E: 76	B: 88	C: 92
B	D: 40	C: 80	A: 88	E: 140
C	B: 80	A: 92	D: 96	E: 100
D	B: 40	A: 50	C: 96	E: 120
E	A: 76	C: 100	D: 120	B: 140

BD	AD	AE	BC	AB
40	50	76	80	88

AC	CD	CE	DE	BE
92	96	100	120	140

A segment is a two-way edge between two cities. After adding/dropping a permanent segment, infer as many new permanent segments as possible to compute the lower bound.

Run TSP BnB to find the optimal tour.

Determine the first three segments xy , uv and pq processed by TSP BnB.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 57 Question Id : 6406531815644 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the lower bound on the cost of the tour as per the TSP BnB algorithm discussed in class?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 17

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

342

Question Number : 58 Question Id : 6406531815645 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the first segment (xy) selected for refining S_0 ?

Enter edge label xy

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: XY

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

BD

DB

Question Number : 59 Question Id : 6406531815646 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cost of the node $\sim xy$ (that permanently excludes segment xy)?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

394

Question Number : 60 Question Id : 6406531815647 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

How many tours are represented by the node $(S0,xy,uv)$?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 61 Question Id : 6406531815648 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cost of the node $(S0,xy,uv,pq)$?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 17

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

346

Sub-Section Number :

5

Sub-Section Id :

640653318020

Question Shuffling Allowed :

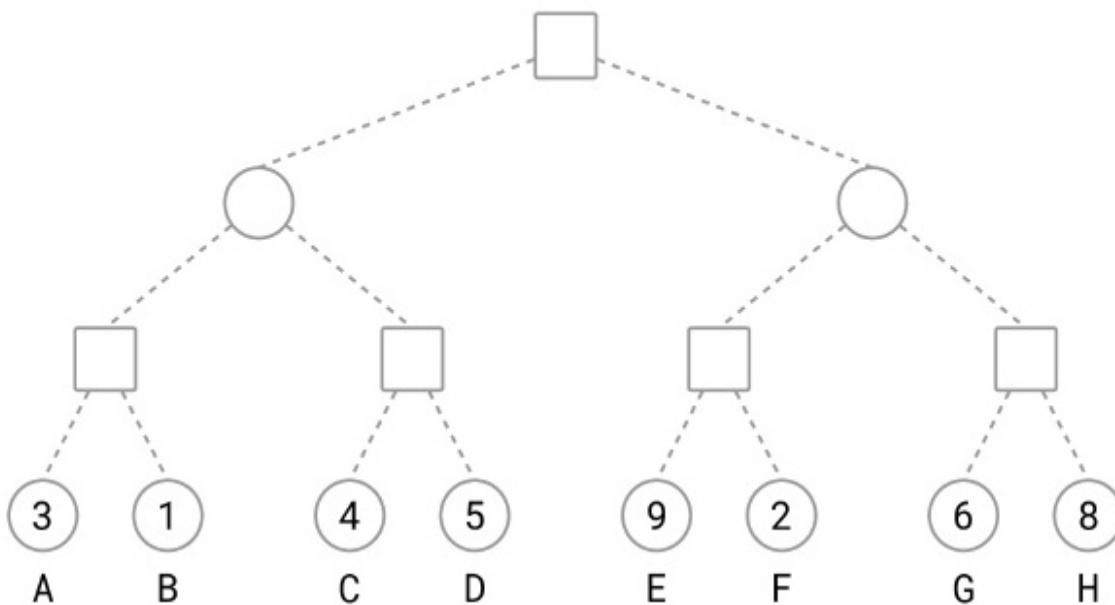
No

Question Id : 6406531815649 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (62 to 68)

Question Label : Comprehension

GAMES



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 62 Question Id : 6406531815650 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes in the best strategy for the MAX player.

Enter node labels as comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

E,H

Question Number : 63 Question Id : 6406531815651 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the total number of strategies available for the MAX player.

Enter an integer

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8

Question Number : 64 Question Id : 6406531815652 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Of all the strategies available for the MAX player, identify the MAX strategy that is most favorable for the MIN player. Enter the horizon nodes of that MAX strategy in the textbox.

Enter node labels as comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

B,C

B,D

Question Number : 65 Question Id : 6406531815653 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes pruned by Alpha-Beta algorithm.

Enter node labels as comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

D

Question Number : 66 Question Id : 6406531815654 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes in the initial cluster formed by SSS* algorithm.

Enter node labels as comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,E,F

Question Number : 67 Question Id : 6406531815655 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Solve the game tree using SSS* algorithm. List the horizon nodes (not in the initial cluster) that are assigned SOLVED status.

Enter node labels as comma separated list in ASCENDING order.
NO SPACES, TABS, DOTS, BRACKETS
OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

G,H

Question Number : 68 Question Id : 6406531815656 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statement(s) about SSS* algorithm.

Options :

6406535902880. ✖ As soon as a MIN node is solved and added to the priority queue, immediately its MAX parent is also marked as solved and added to the priority queue.

6406535902881. ✖ As soon as a last MAX child is solved and added to the priority queue, immediately its MIN parent is also marked as solved and added to the priority queue.

6406535902882. ✔ Only a solved node at the top of the priority queue will determine when its parent will be marked as solved.

6406535902883. ✔ A live terminal node that is at the top of the priority queue will be immediately marked as solved.

Sub-Section Number : 6

Sub-Section Id : 640653318021

Question Shuffling Allowed : No

Question Id : 6406531815657 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (69 to 73)

AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below.

Note: this is the same domain description used in the weekly assignments.

PREDICATES

armEmpty	The arm is not holding any block, it is empty.
holding(X)	The arm is holding X.
onTable(X)	X is on the table.
clear(X)	X has nothing above it, it is clear.
on(X,Y)	X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.

Preconditions: { holding(X) }

Add Effects : { armEmpty, onTable(X) }

Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add Effects : { clear(Y), holding(X) }

Del Effects : { armempty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }

Add Effects : { armEmpty, on(X,Y) }

Del Effects : { holding(X), clear(Y) }

Tie-breaker: When actions are chosen non-deterministically, choose actions that lead to a plan. Throw away the actions that lead to dead-ends or cycles.

Tie-breaker: Treat the goal description, preconditions and effects as lists that are accessed from left to right. When the elements in a list are pushed one by one to a stack, the last element in the list will be at the top of the stack. It has the effect of reversing the list.

A planning problem is given below, find a plan using the operators and predicates defined in the blocks-world domain.

Start: { clear(A), clear(B), holding(A), on(B,C), onTable(C) }

Goal Description: { onTable(B), on(A,B), onTable(C) }

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 69 Question Id : 6406531815658 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the length of the optimal plan? Enter the number of actions in the optimal plan.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 70 Question Id : 6406531815659 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions for the given planning problem?

Options :

6406535902885. ✖ Pickup(A)

6406535902886. ✔ Putdown(A)

6406535902887. ✖ Putdown(B)

6406535902888. ✖ Putdown(C)

6406535902889. ✔ Stack(A,B)